# NATIONAL UNIVERSITY

BANGLADESH




**First Year Syllabus**

**Department of Environment Science**

**Four-Year B.Sc. (Honours) Program**
**Effective from the Session: 2024 – 2025**

**1. Mission**

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world.

**2. Vision**

We expect that on the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.

We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.

By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

**3. Name of the Program:** B.Sc. (Honours) in Environment Science

**4. Program Educational Objectives (PEOs)**

**PEO 1:** To produce graduates with strong theoretical and practical knowledge of Environmental Science aligned with contemporary market and societal demands.

**PEO 2:** To develop graduates capable of conducting interdisciplinary and collaborative research, demonstrating technical competence in environmental monitoring, assessment, and management.

**PEO 3:** To develop graduates who can contribute to the generation of reliable, timely, and relevant environmental data and solutions to support research, planning, and decision-making processes—especially in alignment with the Sustainable Development Goals (SDGs) of Bangladesh.

**PEO 4:** To produce graduates equipped with leadership, teamwork, communication skills, and strong ethical and moral values for responsible and impactful professional engagement in the environmental sector.

**5. Program Learning Outcomes (PLOs):**

**PLO 1: Environmental Knowledge**

Understand and apply core concepts of environmental science—including ecology, climate, biodiversity, pollution, and natural resource management—to solve real-life, academic, and professional environmental problems.

**PLO 2: Problem-Solving Ability**
Recognize, formulate, and solve environmental problems using scientific reasoning, field data, models, and relevant analytical techniques to draw logical and evidence-based conclusions.

**PLO 3: Designing Solutions**
Design and conduct scientific studies to address environmental challenges that meet the specified needs of sustainable living, human welfare, public health, and ecological conservation.

**PLO 4: Research Skills**
Collect, analyze, and interpret environmental data using qualitative and quantitative methods; present results with clarity and draw scientifically sound conclusions.

**PLO 5: Use of Modern Tools**
Utilize modern tools and technologies such as GIS, remote sensing, environmental sensors, and simulation models relevant to environmental monitoring and assessment.

**PLO 6: Environment and Society**
Apply environmental science knowledge to address issues in public health, sustainable development, natural resource use, disaster risk reduction, and community well-being.

**PLO 7: Career and Academic Readiness**
Demonstrate preparedness for careers in environmental consulting, research institutions, teaching, governmental and non-governmental organizations, and qualify for admission to advanced studies in environmental science and related interdisciplinary fields.

**PLO 8: Ethics and Responsibility**
Develop and uphold strong ethical principles and environmental responsibility in personal and professional contexts for the betterment of society and the planet.

**PLO 9: Teamwork and Leadership**
Communicate effectively and work collaboratively in multidisciplinary teams; demonstrate leadership skills to address and manage complex environmental issues.

**PLO 10: Communication Skills**
Convey environmental concepts and research findings effectively through oral presentations, scientific reports, maps, visuals, and digital communication tools to both specialist and non-specialist audiences.

**PLO 11: Project and Time Management**
Plan, manage, and execute environmental fieldwork, laboratory analysis, and research projects efficiently, both individually and in teams, demonstrating responsibility and timely delivery.

**PLO 12: Lifelong Learning**
Engage in continuous learning to keep pace with evolving environmental challenges, scientific advancements, technologies, and policies for personal and professional development.

**6. Mapping PEO with PLO:**

| | PEO-1 | PEO-2 | PEO-3 | PEO-4 |
|---|---|---|---|---|
| PLO-1 | ✓ | | | |
| PLO-2 | ✓ | ✓ | | |
| PLO-3 | ✓ | ✓ | ✓ | |
| PLO-4 | ✓ | ✓ | | |
| PLO-5 | ✓ | ✓ | ✓ | |
| PLO-6 | ✓ | | ✓ | ✓ |
| PLO-7 | ✓ | | ✓ | ✓ |
| PLO-8 | | | ✓ | ✓ |
| PLO-9 | | | | ✓ |
| PLO-10 | | ✓ | ✓ | ✓ |
| PLO-11 | ✓ | | | ✓ |
| PLO-12 | ✓ | ✓ | | ✓ |

## 7. Year wise Courses and marks distribution

# First Year

| Course Code | Course Title | Marks | Credits |
|---|---|---|---|
| 214401 | Introduction to Environmental science | 100 | 4 |
| 214403 | Fundamental of Earth Sciences | 100 | 4 |
| 214405 | Fundamental of Biotic Environment | 100 | 4 |
| 214406 | Laboratory and Field Work on Environmental Sciences-I | 100 | 4 |
| 219901 | History of Bangladesh: Language, Culture and Identity | 100 | 4 |
| 219903 | Information and Communication Technology | 75 | 3 |
| 219904 | Lab: Information and Communication Technology | 25 | 1 |
| | **Any (Two)** | | |
| 212807 | Chemistry-I | 100 | 4 |
| 212808 | Chemistry-I Practical | 50 | 2 |
| 213207 | Introduction to Geography and Environment | 100 | 4 |
| 213208 | Geography and Environment Practical-I | 50 | 2 |
| 213607 | Fundamentals of Statistics | 100 | 4 |
| 213610 | Lab-I: Fundamentals of Statistics | 50 | 2 |
| | **Total =** | **900** | **36** |

| Course Code: | 214401 | Marks:100 | Credits: 4 | Class Hours:60 |
|---|---|---|---|---|
| Course Title: | Introduction to Environmental science | | | |

**Course Objectives**

a) This course aims to introduce students to the fundamental concepts, scope, and interdisciplinary nature of environmental sciences.
b) It develops an understanding of ecosystems, biodiversity, natural resources, and sustainability.
c) Students will explore major environmental issues and their implications through real-world examples and field-based activities.
d) The course also fosters awareness and individual responsibility toward environmental conservation and sustainable living**.**

**Course Learning Outcomes (CLOs)**

| | |
|---|---|
| CLO1 | Knowledge of the general issues of environmental sciences |
| CLO2 | Understanding of the multidisciplinary nature of environmental science |
| CLO3 | Understanding of the basic concept of natural resources and sustainability. |
| CLO4 | Analyze the causes, impacts, and mitigation strategies of various types of environmental pollution and natural disasters. |
| CLO5 | Apply appropriate strategies for conservation of biodiversity and ecosystem management at local, national, and global levels. |
| CLO6 | Develop collaborative and individual action plans to address environmental challenges in Bangladesh and South Asia. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | ✓ | | | | | | |
| CLO2 | ✓ | ✓ | ✓ | | | ✓ | ✓ | | | ✓ | | |
| CLO3 | ✓ | ✓ | | | ✓ | | | | | | | |
| CLO4 | ✓ | ✓ | ✓ | | | | | | | | | |
| CLO5 | ✓ | ✓ | | | ✓ | ✓ | | | | | | |
| CLO6 | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | | | | ✓ |

**Course Contents**

| Module No. | Content | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| 1 | **Basic concept of Environmental Sciences:** History, definition, scope and goal of environmental sciences, its relation to other disciplines of science, state of environmental awareness in society, environmental issues: Greenhouse gases, global warming, acid rain, ozone layer depletion, climate change and sustainable development and action-oriented programs in Bangladesh and South Asia | Lecture (PPT), short videos, group discussion, case-based learning, Q&A | CLO1 & CLO6 |
| 2 | **Multidisciplinary nature of environmental Sciences:** Historical and philosophical basis, concept of Gaia: the interconnectedness of lithosphere, hydrosphere, atmosphere and biosphere, biogeochemical cycles of some important elements | Interactive lecture, concept mapping, collaborative group work, real-world scenario analysis | CLO2 |
| 3 | **Ecosystems, biodiversity and its conservation:** Structure and function of an ecosystem, Ecological succession, Food chains, food webs and ecological pyramids, value of biodiversity, biodiversity at local, national and global levels, Conservation of biodiversity | Multimedia presentation, field-based observation, ecosystem simulation, project-based learning | CLO1 CLO2 & CLO5 |
| 4 | **Natural Resources and sustainability:** Renewable and non-renewable resources, Natural resources and associated problems, Role of an individual in conservation of natural resources, Equitable use of resources for sustainable lifestyles. | Problem-solving tasks, discussion-based lecture, individual action plan preparation, field visits | CLO3 & CLO6 |

| | | | |
|---|---|---|---|
| **5** | **Environmental Pollution:** Causes, effects and control measures of Air Pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution and nuclear hazards, causes and effects of urban and industrial wastes, Role of an individual in prevention of pollution, Concept of Disaster (Floods, earthquakes, cyclones and landslides). | Case study, seminar presentation, group poster-making, role-play, audio-visual demonstrations | CLO4 & CLO6 |

**List of Books**


Botkin, D. B., & Keller, E. A. (1998). *Environmental science: Earth as a living planet* (3rd ed.). John Wiley & Sons.

Manahan, S. E. (2000). *Environmental chemistry* (8th ed.). Lewis Publishers.

Miller, G. T., Jr. (n.d.). *Environmental science*.

| Course Code: | 214403 | Marks:100 | Credits: 4 | Class Hours:60 |
|---|---|---|---|---|
| Course Title: | Fundamental of Earth Sciences | | | |

**Course Objectives**

**a)** To introduce the fundamental concepts of Earth Sciences, including Earth's structure, composition, and its relationship with environmental systems.

**b)** To explore geological processes such as plate tectonics, rock and mineral formation, and paleoenvironmental changes, emphasizing their relevance in understanding Earth's dynamic nature.

**c)** To foster awareness of natural resources, human-environment interactions, and the need for responsible environmental stewardship in the context of Earth's physical systems.

**Course Learning Outcomes (CLOs)**

| | |
|---|---|
| CLO1 | Understand the structure and composition of the Earth system, including its atmosphere, oceans, and space environment. |
| CLO2 | Recognize the importance of Earth's physical environment in the context of a rapidly changing world. |
| CLO3 | Analyze human-environment interactions to understand how human activities are influenced by and impact Earth's systems. |
| CLO4 | Develop informed and responsible citizenship through foundational knowledge of the Earth and its physical processes. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | | | | | | | ✓ |
| CLO2 | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| CLO3 | ✓ | ✓ | | | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ |
| CLO4 | ✓ | | | | | ✓ | | ✓ | | ✓ | | ✓ |

**Course Contents**

| Module No. | Content | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| **1** | **Introduction to Earth Sciences:** Definitions of Earth Sciences (Geological Sciences) and Environmental Science. The scope and subdivisions of geography and Geological Sciences. Relationship between geography, geology and Environmental Sciences. | Lecture with visuals, concept mapping, interactive discussion, short quizzes | CLO1 & CLO2 |
| **2** | **The Universe, Solar System and the Planet Earth:** Big-Bang theory, solar system, different planets, origin of the solar system, Interior of the earth, compositions and physical properties of different layers, geothermal gradient. | Visual presentation, model demonstration, video documentary, group discussio | CLO1 |
| **3** | **Minerals and Rocks:** Mineral's chemistry, definition, physical properties, common minerals - silicate minerals, carbonate, phosphate, and sulfate minerals, ore minerals; Classification, formation, origin and characteristics of different types of rocks. | Lab observation (rock/mineral samples), practical demonstration, chart-based classification, interactive lecture | CLO1 |
| **4** | **Paleoenvironmental study:** fossil, types of preservation, preservation conditions of fossils, types of evolution, reasons of extinction and methods of paleoenvironmental analysis (geologic time, absolute vs relative time, radiometric and relative dating techniques, and geologic time scale). | Fossil visuals/models, timeline building activity, lecture, group analysis task | CLO1 & CLO3 |
| **5** | **Plate Tectonics**: Continental drift, plate tectonics, plate margin, continental collision, cause of plate tectonics, relation amongst plate tectonics, continental crust and mountain building | Animated simulations, case studies, group presentations, mapping exercises | CLO1 & ClO2 |

| | | | |
|---|---|---|---|
| **6** | **Minerals Resources:** Types, amount, and origin of mineral deposits in Bangladesh, prospect and useful mineral substances, natural gas and petroleum reserve in Bengal basin. | Case-based learning, resource mapping, group discussion, data interpretation activities | CLO1 CLO3 & CLO4 |

**List of Books**

Judson, S., & Kauffman, M. E. (1990). *Physical geology*. Prentice-Hall.
Miller, R. W., & Donahue, R. L. (1997). *Soils in our environment*. Prentice-Hall.
Skinner, B. J., & Porter, S. C. (1995). *The dynamic Earth* (3rd ed.). John Wiley & Sons.

| Course Code: | 214405 | Marks:100 | Credits: 4 | Class Hours: 60 |
|---|---|---|---|---|
| Course Title: | Fundamental of Biotic Environment | | | |

**Course Objectives**

**a)** To introduce students to the structure, function, and hierarchical organization of biotic components in the natural environment.
**b)** To explain the origin, evolution, and ecological roles of living organisms within ecosystems and the biosphere.
**c)** To develop students' understanding of biological classification, taxonomy principles, nomenclature rules, and herbarium practices.
**d)** To analyze the ecological and economic significance of microorganisms, plants, and animals in sustaining environmental balance.

**Course Learning Outcomes (CLOs)**

On successful completion of the module, students will be able to:

| | |
|---|---|
| CLO1 | Understand about biotic components of environment. |
| CLO2 | Explain the role and significance of biotic components in environment |
| CLO3 | Demonstrate knowledge of biological classification systems**,** including the principles of taxonomy, nomenclature, and herbarium techniques. |
| CLO4 | Analyze the ecological and economic roles of microorganisms, lower and higher plants, and animals in maintaining environmental balance. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | ✓ | | | | | | ✓ |
| CLO2 | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| CLO3 | ✓ | ✓ | | ✓ | ✓ | | ✓ | | | ✓ | ✓ | ✓ |
| CLO4 | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Course Contents**

| Module No. | Content | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| 1 | **Concept of Biotic Environment:** Biological structure and function of living organism in natural environment; Understanding of hierarchical biotic organization. | interactive lecture, diagram-based explanation, group discussion, brainstorming activity | CLO1 |
| 2 | **Life:** Characteristics, Origin of Life, Evolution of Life; Ecosystem people and Biosphere people. | Visual lecture, storytelling method (origin/evolution), animation/video resources | CLO1, & CLO2 |
| 3 | **Classification of plants**: Artificial, Natural and Phylogenetic systems of classification of plants; Outline of Whittaker's system of classification. | Lecture with classification charts, comparison table, visual taxonomy tree, group analysis | CLO3 |
| 4 | **Taxonomy, Systematic and Herbarium:** Taxonomy and Systematic and their objectives, Principles of nomenclature, author citation, conserved name; Herbarium – herbarium techniques, major herbaria of the world; Bangladesh National Herbarium; Species concept and Taxonomic categories. | Hands-on herbarium activity, specimen identification, field practice, taxonomy demonstration | CLO3 |
| 5 | **Bacteria and Virus:** Archaebacteria and Eubacteria, Gram negative and Gram-positive bacteria, Growth of Bacteria, Mycoplasma, Bacteriophage and its multiplication, Economic and Ecological importance of bacteria and virus. | Microscopic image analysis, lab demo (video), role-play on microbial interaction, case study | CLO4 |
| 6 | **Fungi, Algae and Plankton:** General characteristics, classification, salient features of different classes, economic and ecological importance of fungi, algae and plankton. | Lab visuals or samples, real-life examples, group projects on ecological role, short quiz | CLO4 |
| 7 | Ecological and economic significance of bryophytes and pteridophytes; ecological and economic significance of higher plants and animals. | Multimedia presentation, ecosystem role mapping, comparative discussion, field observation | CLO4 |

| | | | |
|---|---|---|---|
| | | | |
| **8** | General characteristics and classification of animal phyla. | Flashcards, classification charts, short videos, model-based learning, taxonomic tree drawing | CLO3 & CLO4 |

**List of Books**


Ashrafuzzaman, H. (1991). *A text book of plant pathology*. Bangladesh Agricultural Research Council.

Pandey, B. P. (1985). *A text book of botany: Angiosperms*. S. Chand & Company Ltd.

Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (1993). *Microbiology: Concepts and applications*. McGraw-Hill, Inc.

Storer, T. I., & Usinger, R. L. (1979). *General zoology* (3rd ed.). Oxford University Press.

| Course Code: | 214406 | Marks100 | Credits: 4 | Class Hours:60 |
|---|---|---|---|---|
| Course Title: | Laboratory and Field Work on Environmental Sciences-I | | | |

**Section-1: Laboratory Work on Biotic Component of the Environment**

**Course Objectives**

**a)** Develop fundamental laboratory skills for the collection, preservation, and microscopic examination of biological and environmental specimens.
**b)** Apply standard techniques for the identification, characterization, and classification of diverse plant, algal, fungal, bacterial, and aquatic animal groups.
**c)** Integrate field and laboratory practices for accurate documentation, specimen preparation, and long-term preservation in environmental and biodiversity studies.

**Course Learning Outcomes (CLOs)**

On successful completion of the module, students will be able to:

| | |
|---|---|
| CLO1 | Demonstrate proficiency in using microscopes, applying staining techniques, and preparing mounting media for biological specimen analysis. |
| CLO2 | Identify and characterize fungi, algae, aquatic macrophytes, bryophytes, pteridophytes, plankton, bacteria, and selected fish species based on morphological and structural features. |
| CLO3 | Apply proper preservation methods for plant and animal specimens, including herbarium sheet preparation and fauna preservation, in compliance with safety protocols. |
| CLO4 | Compile and present accurate laboratory and field records, integrating observations, drawings, and analytical notes into professional documentation. |

**Mapping CLOs with PLOs**

| | PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 | PLO 8 | PLO 9 | PLO 10 | PLO 11 | PLO 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO 1. | ✓ | | | ✓ | ✓ | | ✓ | ✓ | | | | ✓ |
| CLO 2 | ✓ | ✓ | | ✓ | | ✓ | ✓ | ✓ | | | | ✓ |
| CLO 3 | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | | | ✓ | ✓ |
| CLO 4. | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Course Contents**

| Module No. | Content | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| **1** | Introduction to laboratory; Microscopic Technique; Preservatives, Stains and Mounting Media. | Demonstration and hands-on | CLO1 |
| **2** | Identification and characterization of fungi, algae and aquatic macrophytes | Field collection, microscopy | CLO2 |
| **3** | Study of bryophytes and pteridophytes. | Field study, observation | CLO2 |
| **4** | Demarcation of phytoplankton and zooplankton. | Sampling, microscopic analysis | CLO2 |
| **5** | Study of bacterium: Staining technique. | Staining, microscopic observation | CLO1 & CLO2 |
| **6** | Preparation of herbarium sheet and preservation of fauna. | Specimen preparation, preservation | CLO3 |
| **7** | Identification and characterization of Chondicthis and Osteicthis. | Morphological study, identification | CLO2 |
| **8** | Preparation of field and laboratory note book | Note-taking, record keeping | CLO4 |

**Section-I1: Field Work in Environmental Sciences**

Course Objectives:

**a)** To enable students to observe, identify, and analyze various environmental components and ecosystem interactions through structured field investigation.

**b)** To develop practical skills in field mapping, sampling, and community-based environmental assessment within diverse ecological settings.

**Course Learning Outcomes (CLOs)**

On successful completion of the module, students will be able to:

| | |
|---|---|
| CLO1 | Demonstrate field-based observation and documentation skills by identifying and analyzing various ecosystems, including their biotic and abiotic components, environmental processes, and interspecies relationships. |

| | |
|---|---|
| CLO2 | Apply environmental surveying techniques, including GPS navigation, distance measurement, map plotting, and systematic sampling of soil, water, and biological specimens with accurate labeling and in situ documentation. |
| CLO3 | Evaluate ecosystem functionality and sustainability by interpreting ecological interactions, food chains, geomorphological features, and local resource use through direct observation and stakeholder engagement. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | | | | ✓ | | | |
| CLO2 | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | |
| CLO3 | ✓ | ✓ | | | ✓ | | | | | | | |

**Course Contents:**

| Module No. | Contents | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| 1 | **General Observations:**<br><br>The field visit aims to explore various aspects of the environment and understand ecosystem interactions. Students will observe and identify different ecosystems, their characteristics, and both biotic and abiotic components. Key observations include geomorphology, physiographic conditions, geology, structure, climate, soil, water bodies, and biodiversity of the study area. Attention will be given to how ecosystems function, links between species and their habitats, food chains, ecosystem users, and sustainability. | Field visit, observation checklist, ecosystem sketching, guided group discussion | CLO1, CLO3 |

| | | | |
|---|---|---|---|
| **2** | **Study area:** To understand the natural phenomena of environment, the study area should be such an area where we can meet our all aspects as much as possible. In these considerations, it can be Sitakundo, Chittagong district; Teknaf, Cox's bazaar; Jaflong, Sylhet; Rangamati, Chittagong hilltracts etc. | GPS training, map plotting, sampling demo, informal community interaction | CLO2 & CLO3 |

| Course Code: | 212807 | Marks: 100 | Credits: 4 | Class Hours: 60 |
|---|---|---|---|---|
| Course Title: | Chemistry-I | | | |

**Course Objective**

The course is designed for the student to impart and improve fundamental knowledge and aspects related to understand the fundamental physical, Organic and Inorganic chemistry topics. To develop and extend student's knowledge on state of matter, structure of atom, radiochemistry, chemical bonding, Liquid and Solution, chemical equilibrium, pH, aliphatic and aromatic hydrocarbons.

**Course Learning Outcomes (CLOs)**

After completing this course, students will be able to:

| | |
|---|---|
| CLO1 | Explain fundamental concepts of matter & atomic structure to describe the states of matter and their transitions, apply Avogadro's number, mole concept, and stoichiometry in chemical calculations and analyze atomic models and electronic configurations using quantum numbers and Aufbau principle. |
| CLO2 | Understand chemical bonding & molecular structure to classify types of chemical bonds, predict molecular geometry using VSEPR theory and hybridization and compare valence bond theory (VBT) and molecular orbital theory (MOT). |
| CLO3 | Analyze chemical reactions & equilibrium to differentiate between physical and chemical changes, balance redox reactions and determine oxidation states and apply Le Chatelier's principle to predict equilibrium shifts in reactions. |
| CLO4 | Apply principles of physical chemistry to derive and apply gas laws, calculate colligative properties and interpret Raoult's & Henry's laws and determine pH, buffer capacity, and solubility product (Ksp) in acid-base systems. |
| CLO5 | Classify & characterize hydrocarbons (aliphatic & aromatic) to name hydrocarbons using IUPAC nomenclature, compare reactivity of alkanes, alkenes, and alkynes and explain aromaticity, electrophilic substitution, and substituent effects in benzene derivatives. |

**Mapping of CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | ✓ | | ✓ | | | | | | | ✓ |
| CLO2 | ✓ | ✓ | ✓ | | ✓ | | | | | | | ✓ |
| CLO3 | ✓ | ✓ | ✓ | | ✓ | | | ✓ | | | | ✓ |
| CLO4 | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | | ✓ | | ✓ |
| CLO5 | ✓ | ✓ | ✓ | | ✓ | ✓ | | ✓ | | ✓ | ✓ | ✓ |

**Course contents**

| Module No. | Contents | Teaching-Learning Strategy | CLOs |
|---|---|---|---|
| 1. | **State of Matter in Chemistry:** Nature and classification of matter. Classical states of solid, liquid, gas and Liquid crystals, state of aggregation of matter. Changes of state of matter. Intramolecular and intermolecular force and their role for the determination of state of matter. Modern state as plasma. Avogadro's number with application, Concept of mole: calculation and importance. Stoichiometry. | Lecture, Group Discussion, Assignment | CLO1 |
| 2. | **Structure of atom:** Fundamental particles of atoms, Nucleus and discovery of nucleus, atomic model of Rutherford, Bohr atom model: postulates, limitations and success, Spectrum of atomic hydrogen, Dual nature of electron, Heisenberg uncertainty principle, Quantum numbers, atomic orbitals, Aufbau principle, Pauli exclusion principle, Hund's rule of maximum multiplicity, electronic configuration of atoms. | Lecture, Group Discussion, Assignment | CLO1 |
| 3. | **Radiochemistry:** Atomic number and mass number, atomic mass unit, atomic nucleus, Isotopes, Nuclear stability, natural and artificial radioactivity, half-life and average life of radioelements, radioactive decay, nuclear reactions. | Lecture, Group Discussion, Assignment | CLO1 |
| 4. | **Periodic table:** Periodic law, Periodic table, Periodic properties of the elements such as ionization energies, Electron affinity, Electro negativity, Atomic/ionic radius along a period and down a group, Diagonal relationship. | Lecture, Group Discussion, Assignment | CLO2 |
| 5. | **Chemical bond:** Definition and causes chemical bonds, octet rule and duplet rule, types of chemical bonds: ionic, covalent bond, polar and non-polar bond, Ionic character of covalent compound and covalent character of ionic compounds VSEPR theory, Valence bond theory, Hybridization, $\sigma$ and $\pi$-bonding in compounds, Molecular orbital theory, coordination, metallic, hydrogen bonds. | Lecture, Group Discussion, Assignment | CLO2 |
| 6. | **Chemical Reactions:** Physical change and Chemical change, Methods of bringing about chemical reaction, Redox reactions, oxidizing and reducing agents, oxidation state and oxidation number, redox half reactions, rules for | Lecture, Group Discussion, Assignment | CLO3 |

| | | | |
|---|---|---|---|
| | balancing redox reactions, acid base reactions. | | |
| 7. | **Chemical Equilibrium:** Equilibrium in chemical reactions. Stoichiometry, law of mass action, concentration Verus time curves, equilibrium law and equilibrium constant: Kp, Kc, and Kx, effects of temperature, pressure and concentration changes of equilibrium, principle and applications of Le Chatelier and Braun, reaction quotient, extent of reaction, thermodynamic equilibrium constant. | Lecture, Group Discussion, Assignment | CLO3 |
| 8. | **Gaseous State:** Vapour and gas, measurable properties of gases, determination of gas pressure, the gas laws: derivation, molecular explanation and practical applications, effusion, diffusion, osmosis, ideal and real gases, ideal gas equation with applications, kinetic theory of gas, deviation from ideal behaviour, Van der Walls equation, critical constants. Dalton's law of partial pressure. | Lecture, Group Discussion, Assignment | CLO4 |
| 9. | **Liquids and Solutions:** Liquids and their characteristics, evaporation, measurement vapour pressure, dependence of vapour pressure on temperature variation, boiling, distillation, Types of solution, solubility and solubility curve, units of concentration for solution, Raoult's law, ideal and non-ideal solutions. Henry's law. a brief treatment of colligative properties. | Lecture, Group Discussion, Assignment | CLO4 |
| 10. | **pH and Buffer:** Ostwald dilution law, solubility product with principle and applications, common ion effect, ionic product of water ($K_W$), pH with applications, buffer solution, buffer mechanism, Henderson-Hasselbach equation. Acid-base titrations, Acid-base indicators. Theories of acids and bases, conjugate acids and bases, neutralization reactions, acid-base strength, Salts: Classification and their applications, neutralization reactions. | Lecture, Group Discussion, Assignment | CLO4 |
| 11. | **Aliphatic hydrocarbons:** Organic compounds, homologous series, the causes of diversity of organic compounds, Hydrocarbons: classifications and IUPAC nomenclature. Formation of carbocations, carbanion, free radicals and their stabilities. Sources, properties, general methods of preparation of following hydrocarbons: Alkane, Alkene and Alkynes. Classification of organic compounds based on their functional groups. | Lecture, Group Discussion, Assignment | CLO5 |

| | | | |
|---|---|---|---|
| 12. | **Aromatic hydrocarbons:** Definition and classification, nucleus and side chain, Structure, preparation and uses of benzene. Aromaticity, Electrophilic aromatic substitution reactions with reference to nitration, halogenation, sulphonation and alkylation. Orientation of substituents in benzene ring, activating and deactivating group. | Lecture, Group Discussion, Assignment | CLO5 |

**List of Books**


Bahl, B. S., & Bahl, A. (2024). *Text- Book of Organic Chemistry*. S. Chand & Company Ltd Ram Nagar, New Delhi-1I005S

English, J., H.G Cassidy & Baird, R. I. (1949). *Principles of Organic Chemistry: An Introductory Text in Organic Chemistry*. MeGraw Hill.

Finar, I. L. (1973). *Organic Chemistry, Vol. I*. Pearson Education India.

Griffin, R. W. (2024). *Modern Organic Chemistry*. McGraw Hill.

Morrison, R.T. & Boyd, R. N. (1992). *Organic Chemistry*. Benjamin-Cummings Pub Co

Muny, J. Mc. (2010). *Organic Chemistry.* Thomson Broks-Coole.

Roberts, J. D., & Caserio, M. C. (1977). *Basic Principles of Organic Chemistry*. W. A. Benjamin, Inc., Menlo Park, CA.

Skoog, D. A., & West, M. D. (2022). *Fundamentals of analytical chemistry*. Publisher, Cengage Learning.

Solomons, T. W. G. (2008). *Fundamental of Organic Chemistry*. Wiley.

| Course Code: | 212808 | Marks: 50 | Credits: 2 | Class Hours: 30 |
|---|---|---|---|---|
| Course Title: | Chemistry-I Practical | | | |

**Course Objectives**: To gain a fundamental knowledge on laboratory safety policy, some inorganic salts and radical analysis, preparation of solution with standerization, titration and precipitation.

**Course Learning Outcomes (CLOs)**

After completing this course, students will be able to:

| | |
|---|---|
| CLO1 | Discuss about the general laboratory safety policy, rules and regulations and chemical management process. |
| CLO2 | Explain the applications of apparatus and reagents in various experiments, different laboratory glassware and chemicals. |
| CLO3 | Prepare stander solution and pH- neutralization curves |
| CLO4 | Qualitative analysis of mixtures of inorganic salts consisting of up to five different radicals |
| CLO5 | Explain to determination of $Fe^{2+}$, copper (II), nickel as Ni (HDMG) 2 complex 7 using difference methods. |

**Mapping of CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | ✓ | ✓ | | ✓ | | | | ✓ |
| CLO2 | ✓ | ✓ | ✓ | | ✓ | ✓ | | | | | ✓ | ✓ |
| CLO3 | ✓ | ✓ | ✓ | | ✓ | | | | | | | ✓ |
| CLO4 | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ |
| CLO5 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | ✓ | ✓ | ✓ |

**Course contents**

| Module No. | Contents | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| 1 | **Safety:** Introduction of chemical, equipment's and safety in the laboratory. | Lecture, Experimental, Assignment | 1-5 |
| 2 | **Preparation:** Preparation of $FeSO_4.7H_2O$ Mohr's salt and potash alum. | Lecture, Experimental, Assignment | 1-5 |
| 3 | **Separation and identification**: Separation and identification of four radicals from a mixture of anions and cations The cations are $Pb^{2+}$, $Cu^{2+}$, $Cd^{2+}$, $Al^{3+}$, $Fe^{2+}$, $Fe^{3+}$, $Co^{2+}$, $Ni^{2+}$, $Zn^{2+}$ $Ca^{2+}$, $Ba^{2+}$, $Na^{+}$, $K^{+}$, and $NH_4^{+}$, the anions are $NO_3^{-}$, $CO_3^{2-}$, $S^{2-}$, | Lecture, Group Discussion, Experimental, Assignment | 1-5 |

| | | | |
|---|---|---|---|
| | $SO_4^{2-}$, $Cl^-$ , $Br^-$ and $I^-$. | | |
| **4** | **Standardization:** Standardization of NaOH solution using standard oxalic acid solution. | Lecture, Experimental, Assignment | 1-5 |
| **5** | **Determination:** Determination of $Fe^{2+}$ using standard permanganate solution. | Lecture, Experimental, Assignment | 1-5 |
| **6** | **Iodometric method:** Iodometric determination of copper (II) using standard $Na_2SO_3$ solution. | Lecture, Group Discussion, Experimental, Assignment | 1-5 |
| **7** | **Gravimetric Method:** Gravimetric determination of nickel as Ni $(HDMG)_2$ complex. | Lecture, Group Discussion, Experiment, Assignment | 1-5 |
| **8** | **pH-Neutralization Curves:** Determination of the pH-neutralization curves of a strong acid by a strong base. | Lecture, Experiment, Assignment | 1-5 |

**List of Books**


Alexeyev, V. (2024). *Qualitative Analysis*. Mir Publishers.

Braun, D. R. (2024). *Introduction to Chemical Analysis*. McGraw Hill International.

Christian, G. D. (2023). *Analytical chemistry*. John Wiley & Sons.

Vogel, A. I. (2017). *A Text-Book of Macro and Semimicro Qualitative Inorganic Analysis*. Longmans, Green and Co. Ltd.

Welcher, F. J., & Hahn, R. B. (2024). *Semimicro Qualitative Analysis*. D. Van Nostrand Co. Inc.

| Course Code: | 213207 | Marks: 100 | Credits: 4 | Class Hours:60 |
|---|---|---|---|---|
| Course Title: | Introduction to Geography and Environment | | | |

**Course Description**

This course provides the basic concept of geography and earth system science with the definition, meaning, nature and scope. The content of the course is also described the astronomical aspects of the earth. After completion of this course students will learn about the universe, galaxy, solar system and the earth along with the understanding of latitude, longitude and global timing.

**Course Learning Outcomes (CLOs)**

| | |
|---|---|
| CLO1 | Able to understand the introduction of geography and environment, and their relations giving emphasis on astronomical knowledge. |
| CLO2 | Able to describe the creation of Universe, formation of galaxy, Solar System and differences between Earth and other planets in the Solar System |
| CLO3 | Able to explain earth's co-ordination system, global calendar system and its application in the real life. |
| CLO4 | Able to analyze the Earth's movement (rotation and revolution) and its effects on time, day length, seasons, and illumination. |
| CLO5 | Able to interpret time zone differences and calendar systems from different cultural and geographic perspectives. |
| CLO6 | Able to evaluate the interrelationships between environment, society, and geography with emphasis on human-environment interaction. |
| CLO7 | Able to collaborate in group activities and demonstrate critical thinking through map reading, timeline analysis, and concept mapping. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | | ✓ | | | | | |
| CLO2 | ✓ | ✓ | ✓ | | | ✓ | ✓ | | ✓ | ✓ | | |
| CLO3 | ✓ | ✓ | | | ✓ | | | | | | | |
| CLO4 | ✓ | ✓ | ✓ | | | | | | | | | |
| CLO5 | ✓ | ✓ | | | ✓ | ✓ | | | | | | |
| CLO6 | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | | | | ✓ |
| CLO7 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Course Contents**

| Module No. | Contents | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| **1** | **Geography and Environment:** Meaning, fundamental concepts, scope, nature and branches. The relationship between geography and astronomy | Lecture (PPT), short video clips, concept mapping, group discussion, Q&A, think–pair–share | CLO1 |
| **2** | **Universe:** Big-Bang theory, creation of primary elements, formation of galaxy, and stars. | Multimedia lecture, group activity on cosmic timeline, guided Q&A | CLO2 |
| **3** | **Solar System:** Planets and Satellites, Sun, Earth and Moon. | PPT-based lecture, discussion, solar system model interpretation, visual quiz | CLO2 |
| **4** | **Earth System:** Form and shape of the earth including oblate and spheroid curvature of the earth. Common centre of gravity. Polar axis and equatorial axis. | Diagram demonstration, 3D globe activity, interactive lecture, problem-solving | CLO3 |
| **5** | **Earth's Circumference and Area:** Longitude and Latitude determination. Prime meridian. Length of latitudes and longitudes. Great circle. Rhumb line. | Practical mapping task, group exercise, case examples, peer feedback discussion | CLO3 & CLO7 |

| | | | |
|---|---|---|---|
| 6 | **Rotation and Revolution:** Illumination of the globe. Circle of illumination. Almanac and analema. Length of day and night, and altitude of the sun. Solstices and equinoxes. Twilight - types, determination. | Lecture with animation, diagram-based demonstration, almanac analysis, class discussion | CLO4 & CLO7 |
| 7 | **Time:** Longitude and time. Local and standard time, division of the earth according to standard time. Standard time of USA, India and Bangladesh. International Date Line. Calendar- Astronomical, Jewish, Mohammedan, Roman, Julian, Gregorian and World.<br><br>**Time Zone Calculation:** Mathematics of Time Zone Calculation, International and Local Time Zones. | Timeline construction, comparative charting, calendar workshop, case-based learning | CLO5 & CLO7 |
| 8 | **Environment:** Meaning, structure, types and components of Environment; Geography and Environment; Environment and Society, Ecology, Sustainability. | Interactive lecture with PPT, case study analysis on human-environment interaction | CLO6 |

**List of Books**


Adhikari, S. (2011). *Fundamentals of geographical thought*. Chaitanya Publishing House, Allahabad, India.

Arny, T., & Schneider, S. E. (1994). *Explorations: An introduction to astronomy*. Mosby.

Berry, B. J. L. (1973). *A paradigm for modern geography*. In R. J. Chorley (ed.), *Directions in geography*. Methuen, London.

Broek, J. O. (1965). *Geography: Its scope and spirit*. Charles E. Merrill, Columbus, Ohio.

Branley, F. M. (1975). *Astronomy.* Thomas Y. Crowell & Co., New York.

Singh, S. (2001). *Environmental geography*. Prayag Pustak Bhawan, India.

Strahler, A. H., & Strahler, A. N. (2006). *Introducing physical geography*. Wiley.

| Course Code: | 213208 | Marks: 50 | Credits: 2 | Class Hours: 30 |
|---|---|---|---|---|
| Course Title: | Geography and Environment Practical-I | | | |

**Course Description**

This course offers practical, map-based lab activities focused on fundamental and advanced cartographic techniques. Students will learn about various types of maps, their features, and applications, including map construction, transformation, scale use, enlargement, and reduction. The course emphasizes thematic mapping—qualitative, quantitative, chorochromatic, and choreographic—as well as the interpretation of topographic, cadastral maps, and aerial photographs. Skills will be developed in drawing cross-profiles from contour lines, identifying physical features, applying interpolation techniques, generating isopleths from spot heights, using vertical exaggeration, and conducting inter-visibility analysis.

**Course Learning Outcomes (CLOs)**

| | |
|---|---|
| CLO1 | Able to define the fundamental concepts and basic principles of maps and map making techniques. |
| CLO2 | Able to acquire knowledge of basic cartographic techniques and tools, including map scales, and the construction, transformation, and reproduction of maps using topographic maps, cadastral maps, and aerial photographs |
| CLO3 | Able to design/construct map scale for different type of thematic maps. |
| CLO4 | Able to be skilled for map design, preparation, reproduction, reduction and enlargement of map. |

**Mapping CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | | | ✓ | | | | |
| CLO2 | ✓ | ✓ | ✓ | | ✓ | | | ✓ | ✓ | ✓ | ✓ | |
| CLO3 | ✓ | ✓ | ✓ | | ✓ | | | ✓ | ✓ | | | ✓ |
| CLO4 | ✓ | ✓ | ✓ | | ✓ | ✓ | | ✓ | | | ✓ | ✓ |

**Course Contents**

| Module No. | Contents | Teaching-Learning Strategies | CLOs |
|---|---|---|---|
| **1** | Construction of map scale (simple, comparative, and diagonal). | Lecture with Demonstration, Guided Drawing Practice, Q/A Session, Class Discussion | CLO1 CLO2 & CLO3 |
| **2** | Transformation of map scale and measurement of area | Group Work, Hands-on Exercise, Field Demonstration, Use of Graph Paper/Planimeter | CLO2 & CLO4 |
| **3** | Drawing of thematic maps. Qualitative, quantitative, chorochromatic, and choreographic maps. | Field Visit, Practical Mapping Session, Role Play (map interpretation), Simulation Tools (GIS-based if available) | CLO1 & CLO3 |
| **4** | Map design, preparation, reproduction, reduction and enlargement of map. | Seminar (on Map Errors & Best Practices), Hands-on Practice, Drawing Lab Sessions, Use of Photocopying & Scanning Tools | CLO3 & CLO4 |
| **5** | Getting familiar with topographic maps, cadastral maps, aerial photographs of both from Bangladesh and from other countries | Lectures with Visual Aids (maps/photographs), Hands-on Lab Sessions, Class Discussions, Pair Work (Map Reading) | |

**List of Books**


Monkhouse, F. J., & Wilkinson, H. R. (1976). *Maps and diagrams: Their compilation and construction*. Methuen, London.

Raisz, E. (1948). *General cartography*. McGraw-Hill, New York.

Robinson, A. H. (1978). *Elements of cartography*. John Wiley & Sons, New York.

Singh, R. L. (1979). *Elements of practical geography*. Kalyani Publisher, New Delhi.

| Course Code: | 213607 | Marks:100 | Credits: 4 | Class Hours: 60 |
|---|---|---|---|---|
| Course Title: | Fundamentals of Statistics | | | |

**Course Objectives**

To be able to understand the nature, characteristics, scope, application and abuse of statistics. To make familiar with data, nature of data, how to process and condense the data, sources of data and graphical presentation of data, and to apply appropriate statistical tools and techniques to analyze the data. To acquaint students with necessary skills for solving probability related problems using appropriate laws. To provide knowledge on time series and statistical indices.

**Course Learning Outcomes (CLOs)**

After going through the course, students will be able to learn the following outcomes

| | |
|---|---|
| CLO1 | Understand the nature, characteristics, scope, application and abuse of statistics. Knowledge on sources of data and how to process, condense and presentation of data. Understand different characteristics of statistical data such as measures of location, dispersion, moments, skewness, kurtosis and their properties. |
| CLO2 | Understand the relationship between variables such as simple correlation, rank correlation, correlation ratio, simple regression analysis. standard error of estimators of regression coefficients & their properties and fitting of regression lines. |
| CLO3 | Comprehend different approaches of defining probability and useful laws of probability to solve problems. Also learn some commonly used probability distributions. |
| CLO4 | Gain knowledge about meaning and application of statistical indices. |
| CLO5 | Identify the pattern and trends and isolate the influencing factors of the time series data for future planning and control. |
| CLO6 | Understand the concept of numerical methods, including interpolation and use of numerical methods in application to real problems. |

**Mapping of CLOs with PLOs**

| | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CLO1 | ✓ | ✓ | | | | | | | | | | ✓ |
| CLO2 | ✓ | ✓ | ✓ | | | | | | | | | ✓ |
| CLO3 | ✓ | ✓ | ✓ | | | | | | | | | |
| CLO4 | ✓ | ✓ | | | | | | | | | | |
| CLO5 | ✓ | ✓ | ✓ | | | | | | | | | ✓ |
| CLO6 | ✓ | ✓ | | | | | | | | | | ✓ |

## Course Contents

| Module No. | Contents | Teaching Learning Strategies | CLOs |
|---|---|---|---|
| **1** | Descriptive Statistics: Statistics–Its nature and some important uses, Qualitative and quantitative data, Classification, Tabulation and frequency distribution, Graphical representation of data, Measures of location, Measures of Dispersion, Skewness and Kurtosis, Mathematical relationship among different measures of location, dispersion, Skewness and kurtosis. | Lecture, Assignment | CLO1 |
| **2** | Bivariate Data: Correlation coefficient, Correlation analysis, The purpose and uses of regression analysis, Simple regression and methods of least squares and estimation of parameters, Correlation ratio, Rank correlation, Partial and multiple correlation. | Lecture, Assignment | CLO2 |
| **3** | Elementary Probability: Meaning of Probability, Classical and empirical definitions of Probability, Axiomatic approach of defining probability, Event, Sample space and simple problems on probability, Addition rule, Conditional probability, Multiplication rule and Bayes theorems, The concept of a random variables, Probability function and probability density function, Joint probability function. Marginal and conditional distributions, Statistical independence, expected value and related theorems, Moment generating function, Common probability distributions, Binomial, Poisson and Normal. | Lecture, Group Discussion, Assignment | CLO3 |
| **4** | Index Number: Concept of an index number and problems in the construction of index number, Types of indices (Price, Quantity, Value and cost of living indices) and their uses, Tests for index numbers. | Lecture, Group Discussion, Workshop and Assignment | CLO4 |
| **6** | Time Series analysis: Elements of time-series analysis, Measurement of trend by moving average, By least square method, Trend curve, Determination of seasonal indices, | Lecture, Group Discussion, and Assignment | CLO5 |

| | | | |
|---|---|---|---|
| | Cyclical movements. | | |
| **7** | Numerical Mathematics: Differences of a polynomial, Finite difference operator, Difference table, Newton's formula and starling's central difference formula, Inverse interpolation, Numerical integration. | Lecture, Group Discussion, Workshop and Assignment | CLO6 |

**List of Books**


Gupta, S.C. & Kapoor, V.K. (1994). Fundamentals of *Applied Statistics*. Sultan Chand & Sons.

Hoel, P.G. (1971). *Introduction to Statistics (4th ed.*). Wiley and Sons

Islam, M.N. (2015). *An introduction to Statistics and Probability (4th ed.).* Mullick & Brothers.

Jalil, M. A., & Ferdous R. (1999). *Basic statistics: Methods and Applications*, Robi Publications.

Mostafa, M.G. (1989). *Method of Statistics (4th ed.).* Karim press and Publications.

Shil, R. N., & Debnuth, S.C. (2016). *An introduction to the theory of Statistics*. Star Publications.

Simpson, G., & Kapka, F. *Basic Statistics.* Oxford Ibh Publishing Co.

Weiss, N. A., & Weiss, C. A. (2012). *Introductory statistics*. Pearson Education.


| **Course Code:** | **213610** | **Marks:50** | **Credits: 2** | **Class Hours: 30** |
|---|---|---|---|---|
| **Course Title:** | **Lab-1: Fundamentals of Statistics** | | | |

Data condensation and tabulation. Formation of frequency distribution from both qualitative and quantitative data. Construction of bivariate table. Graphical representation of data. Measures of location and dispersion, Calculation of moments, Measures of skewness and kurtosis. Simple correlation coefficient and fitting of regression lines. Computation of rank correlation coefficient. Fitting of Binomial, Normal and Poisson's distributions, Finding trend values and seasonal variation from time series data by different methods, Calculation of Index numbers and test of index number, Use of Newton's forward and backward formula, Solution of numerical integration.

| Course Code: | 219901 | Marks: 100 | Credits: 4 | Class Hours: 60 |
|---|---|---|---|---|
| Course Title: | History of Bangladesh: Language, Culture and Identity | | | |

**Course Objective**

This course is designed to help undergraduate students from diverse academic backgrounds develop a comprehensive and nuanced understanding of the historical development of Bangladesh, with a particular focus on its language, culture, and identity. By critically examining a variety of historical events, socio-political movements, and cultural shifts from ancient times to the contemporary period, students will have the opportunity to trace the evolution of the Bengali nation. The course aims to foster informed citizenship through an exploration of the Liberation War, identity formation processes, cultural heritage, the lives and contributions of key political figures, and the roles of ethnic minorities. Ultimately, the objective is to equip learners with the intellectual tools necessary to contextualize current national debates and to recognize the role of youth and globalization in shaping the future of Bangladesh.

**Course Learning Outcomes (CLOs)**

At the end of the course, learners will be able to:

CLO 1: Recall key historical events, dates, movements, personalities, and cultural developments that contributed to the emergence of Bangladesh from the pre-colonial period to the present day.

CLO 2: Demonstrate an understanding of the socio-political, linguistic, and cultural influences that shaped Bengali identity and nationalism, emphasizing the Language Movement, Liberation War, and subsequent democratic uprisings.

CLO 3: Apply their historical and cultural understanding to interpret contemporary national issues and debates relating to identity, political reform, and cultural transformation in Bangladesh.

CLO 4: Analyze how historical events, political ideologies, and social changes are interconnected in shaping the current socio-political landscape of Bangladesh, with a particular emphasis on the roles of movements, minorities, and the youth.

**Course Contents**

| Course contents | Teaching Learning Strategy | CLOs |
|---|---|---|
| **1. Pre-colonial Era**<br>• Life and Culture of the People in Ancient Bengal | Interactive lectures | CLO 1-2 |

| | | |
|---|---|---|
| • Bengal Under Muslim Rulers: Society, Culture, and Religion<br>• Role of the Sufis in Preaching Islam and Impacts of<br>• Sufism on the Bengali Society<br>• Bengali Society and Culture in the Writings of Foreigners | | |
| **2. Colonial Era (18th and 19th Century)**<br>• The Battle of Plassey (1757) and the Beginning of British Colonialism<br>• Bengal Renaissance<br>• Reforms in Hindu Society<br>• New Forms in Bengali Literature and Culture<br>• The Muslim Response to Western Education | Interactive lectures, reading, and assignments | CLO 1-2 |
| **3. Colonial Era (First Half of the 20th Century)**<br>• Partition of Bengal (1905)<br>• Hindu-Muslim Disagreements<br>• Formation of the All-India Muslim League (AIML)<br>• Muslim Shahitya Samaj<br>• Buddhir Mukti Andolan: The Urge for Rational Thinking in Bengali Muslim Society<br>• Growth of Religion-based Identity<br>• Politics of Hindutva and the Two-Nation Theory<br>• Spread of Communalism in Society<br>• The Partition of India and Bengal | Interactive lectures and group discussions | CLO 2-3 |
| **4. Post-Partition Era (1947-1971)**<br>• Language Movement<br>• Political, Economic, and Cultural Aspects<br>• Growth of Vernacular Nationalism<br>• Cultural Activism | Interactive lectures and group discussion | CLO 2-3 |
| **5. Changing Bengali Identity**<br>• The Evolution of Bengali Identity in the Context of Language, Culture, and Religion<br>• From Ancient Times to the Present<br>• The Role of the Bengali Language in Shaping Identity<br>• The Language Movement of 1952 and Its Long-term Impact on National<br>• Consciousness<br>• The Influence of Religion on Bengali Identity<br>• The Impact of Socio-political Movements on Identity Formation | Interactive lectures and thematic assignments | CLO 2-3 |

| | | |
|---|---|---|
| • The Liberation War of 1971<br>• Contemporary Debates on Bengali Identity<br>• The Role of Youth in Redefining Identity<br>• The Influence of Globalization | | |
| **6. Liberation Movement of 1971 and Mass Uprising till 2024**<br>• The Political and Economic Exploitation of East Pakistan by West Pakistan<br>• Six (6) Points Movement, Uprising of 1969<br>• The 1970 General Elections<br>• The Non-cooperation Movement and the Declaration of Independence<br>• The Role of AK Fazlul Haque, Huseyn Shaheed Suhrawardy, Maulana Bhashani, Sheikh Mujibur Rahman and Ziaur Rahman<br>• The Liberation War of 1971<br>• Genocide and Resistance<br>• The Role of the Mukti Bahini<br>• Post-independence Challenges<br>• Nation-building<br>• Political Instability<br>• Mass Uprisings and Democratic Movements<br>• The Anti-autocracy Movement of the 1980s<br>• The 1990s Movement for Democracy and Afterwards<br>• The 2024 Movements for Political and Social Reforms | Interactive lectures, group discussions, and thematic assignments | CLO 1<br>CLO 3-4 |
| **7. History of Other Ethnic Groups**<br>• The Indigenous Communities of Bangladesh<br>• Historical Presence<br>• Cultural Practices<br>• Land Rights and Cultural Assimilation<br>• Contributions and Sacrifices<br>• Political Representation of Ethnic Minorities | Interactive lectures and group discussions | CLO 4 |
| **8. Cultural Heritage and Modern Transformations**<br>• The Evolution of Bengali Culture<br>• From Ancient Traditions to Modern Expressions<br>• The Role of Literature, Music, and Art in Shaping Bengali Identity<br>• Contributions of Rabindranath Tagore, Kazi Nazrul Islam, and Other Cultural Icons<br>• The Impact of Globalization on Bengali Culture | Interactive lectures, documentary screening, and thematic assignments | CLO 1-4 |

| | | |
|---|---|---|
| • The Revival of Traditional Arts<br>• The Role of Youth in Cultural Innovation Urbanization<br>• Environmental Changes<br>• The Commodification of Culture | | |

**List of Books**


Ahmed, M. (1979). *Bangladesh: The constitutional quest for autonomy*, Dhaka: University Press Limited.

Ahmed, R. (1981). *The Bengal Muslims 1871-1906: A quest for identity.* Oxford University Press.

Alavi, H. (1972). *The state in post-colonial societies: Pakistan and Bangladesh.* In K. Gough & 11. P. Sharma (Eds.), Imperialism and revolution in South Asia (pp. 145-178). New York, NY: Monthly Review Press.

Bernier, F. (2023). *Travels in the Mogul Empire: A.D. 1656-1668.* Chennai: Atlantic Publishers.

Bleie, T. (2005). *Tribal peoples, nationalism, and the human rights challenge.* The Adivasis of Bangladesh. University Press Limited.

Bosc, S. (2011). *Dead reckoning Memories of the 1971 Bangladesh war*. Hurst & Company.

Eaton, R. M. (1996). *The rise of Islam and the Bengal frontier. 1204-1760.* Berkeley: University of California Press.

Gilmour, D. (2019), *The British in India: Three centuries of ambition and experience*. London. Penguin.

Habib, L. (1982). *Cambridge economic history of India. Cambridge.* Cambridge University Press.

Halim, S., Amanullah, A. S. M., & Nasir, R. 1. (Eds.). (2024). *Society and sociology in Bangladesh*: *A South Asian perspective.* The University Press Limited.

Hashmi, T. (2021). *Fifty Years of Bangladesh, 1971-2021 Crises of Culture, Development, Governance and Identity.* Switzerland: Palgrave Macmillan.

Huq, M. E. (1975). *History of Sufism in Bengal. Dhaka.* Bangladesh Asiatic Society

Husain, 1. (2014). *Karl Marx on India.* New Delhi: Tulika Books.

Jahan, R. (1972). *Pakistan: Failure in national integration.* New York, NY: Columbia University Press.

Maniruzzaman, T. (1988*). The Bangladesh revolution and its aftermath (2nd ed.)*. Dhaka: University Press Limited. (Original work published 19801

Mascarenhas, A. (1986). *Bangladesh: A legacy of blood. London:* Hodder and Stoughton.

Rashid, H. O. (2015). *The foreshadowing of Bangladesh: Bengal Muslim League and Muslim politics* (3rd ed.). Dhaka: University Press Limited.

Roy, A. (1984). *The Islamic syncretistic tradition in Bengal. Princeton.* NJ: Princeton University Press.

Sen, A. (2006). *Identity and violence: The illusion of destiny.* W. W. Norton & Company.

Umar, B. (2022). *The Emergency of Bangladesh: A History of East Pakistan.* Dhaka: Bangla Gobeshona.

| Course Code | 219903 | Marks: 75 | Credits: 3 |
|---|---|---|---|
| Course Title: | Information and Communication Technology | | |

**Course Objectives:**

The main objective of the course is to develop students' understanding and skills in using, managing, and applying technology to solve problems and enhance various aspects of life and work. Key areas of focus include understanding ICT systems, software, hardware, networks, and their applications.

**Course Learning Outcomes:** The student will be able to:

| | |
|---|---|
| CLO 1 | Explain foundational ICT concepts, including the information processing cycle. |
| CLO 2 | Operate standard computer hardware and software systems effectively. |
| CLO 3 | Use office productivity tools (Word, Excel, PowerPoint) for academic and professional tasks. |
| CLO 4 | Apply safe internet practices and use internet tools for communication and information retrieval. |
| CLO 5 | Understand and explain the concepts of Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML) |
| CLO 6 | Understand the concept of free-lancing |

## Course Contents:

| Unit | Specific Objectives | Content | Teaching and Learning Approach |
|---|---|---|---|
| **Unit 1:** Introduction to Information and Communications Technology (ICT) and Computer System | The student will be able to:<br>• Explain the concept of ICT and its related terminologies.<br>• Describe the information processing cycle.<br>• Analyze the impact of ICT on educational, social and economic | Definition of ICT, basic concepts and terminologies. Data, Information<br>Application of ICT, Advantages and Disadvantages of ICT, Impact of ICT Career opportunities in ICT education<br>Information Processing Cycle. | Classroom Lectures |

| | | | |
|---|---|---|---|
| | development.<br>• Identify career opportunities in ICT education.<br>• Differentiate among the classes of computers and the usages of them.<br>• Identify the vital components of the Systems Unit. | Classification of Computers<br>The Vital Components of the Systems Unit. | |
| **Unit 2:**<br>Computer Hardware and Software | The student will be able to:<br>• Describe the categories of Computer Hardware.<br>• Describe the commonly used Input and Output devices<br>• Identify the main processing devices, storage devices and media. Identify the main communication devices.<br>• Identify types of Software packages.<br>• Distinguish between an Operating System and Application software.<br>• Identify different types, examples and uses of Operating Systems and Application software<br>• Distinguish between Open Source and Proprietary Software. | Categories of Computer Hardware: Input devices, Processing devices, Output devices, Storage devices<br>Communication devices,<br>Main Processing Devices: The Processor, Control Unit and Arithmetic and Logic Unit<br>Software Packages,<br>Operating Systems, Types and uses of Operating Systems, Types and uses Application Software: Educational software, Games software, Graphics software<br>Browsers: Internet explorer, Google chrome, Mozilla Firefox, Opera, Internet explorer, Mozilla Firefox; Proprietary and Open Source Software | Classroom lectures and Lab |
| **Unit 3:**<br>Introduction to Word Processing Application | • The student will be able to:<br>• Identify Word Processing packages.<br>• Create and save a document using the Word Processor.<br>• Format a Word document using formatting tools. | Word Processing Packages and Their Uses. Introduction of Microsoft Word Software.<br>File creation and saving: New, Save, Open, Save As.<br>Editing: Copy/Cut, Paste, Undo, Redo, Find, Replace, Clipboard, Sub-Headings And Paragraphs, Text Correction, | Classroom Lectures, Lab, and Hands-on Practice |

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| | Demonstrate the ability to perform collaborative editing.<br>• Insert tables in a Word Processing document.<br>• Insert symbols and pictures in Word Processing documents.<br>• Use layout techniques in document creation.<br>• Inserting headers and footers.<br>• Print documents using the various print options. | Wrapping Text, Text Orientation<br><br>**Formatting:**<br>Font: Style, Size, Color, Bold, Underline, Italic, Superscript, Subscript, Shadow, Strikeout.<br>Paragraph: Alignment, Bullet & Numbering, Indent, Line Spacing, Table Border<br><br>**Insert:** Table, Picture, Shape, Smart Art, Chart, Header, Footer, Page Number, WordArt, Drop Cap, Symbols, Equations, Hyperlink, Comment.<br><br>**Page Design:** watermark, page border<br>**Layout:** Page setup, print | |
| **Unit 4:** Spreadsheet Application | • The student will be able to:<br>• Identify Spreadsheet Packages.<br>• Explain the importance of the Spreadsheet application in data management.<br>• Explain related concepts and terminologies in the Spreadsheet.<br>• Identify features in the Spreadsheet application window.<br>• Create and save a Workbook.<br>• Construct and insert simple formulae and functions.<br>• Format the worksheet using formatting tools.<br>• Printing a worksheet. | Spreadsheet Packages, e.g. Excel. Importance of Spreadsheet application in Data Management, Related Concepts and Terminologies (e.g., cell(s), rows, columns, worksheet, workbook)<br><br>Excel Tool Bars: Formatting Bar, Standard Bar, Formula Bar.<br><br>Types of Data and Their Uses (e.g., number, date, text, currency), Creating and Saving a Workbook, Formula and Functions, Sorting, filtering, validation, creating pivot table.<br><br>Formatting Worksheet Using Formatting Tools.<br><br>Draw Graph/chart, Printing Worksheet. | Classroom Lectures, Lab and Hands-on Practice |

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| **Unit 5:** Presentation Application | • The student will be able to:<br>• State the importance of the Presentation application.<br>• Identify the commonly used features of a Presentation application while prepare a presentation.<br>• Create and save presentations using a template, Add new slide(s). Edit text, Format text, Insert objects, images and pictures, Run slide show, Apply transition, animation effects to slides<br>• Select the print option for printing.<br>• Prepare a presentation on a selected topic and present it. | Presentation Applications Packages, Devices used for Presenting, Importance of Presentation Application, Principles for Designing Presentations, Terminologies in Presentation<br><br>Application (eg, , Slide transitions, Slide show, etc.),<br><br>Commonly Used Features of Presentation Application Window: Toolbars, Different presentation view modes<br><br>Prepare a Presentation: Slide creating, editing, theme, background, inserting elements (image, audio, video hyperlink, table, smart art, graph, etc.) and formatting slides, Slide Layout, slide Show, slide transition, animation Effects.<br><br>Selection of Print Option: Entire presentation, Specific slides, Handouts, Notes pages, outline view of slides. | Classroom Lectures, Lab and Hands-on Practice |
| **Unit 6:** Privacy and Security | The student will be able to:<br><br>• Understand the basics of digital security<br>• Use some security tools.<br>• Understanding digital ethics. | Introduction to Information Security, cybercrime, DoS and DDoS Attack, Key Management, Digital Signature and Certifications, privacy, Data Security, Vulnerability, Threat and Risk, Malware, Social Engineering, Hacking, Plagiarism, Fishing, Software Piracy, Worms and Viruses, Spam, Adware, Malware, Spyware, Antivirus Software<br><br>Ethics in the digital world | Classroom Lectures, Lab and Hands-on Practice |
| **Unit 7:** Using The Internet to Communicate | • The student will be able to: | Internet, Intranet, Extranet, IP Address, Masking, MAC Address, Internet Services, | Classroom Lectures, Lab and Hands-on |

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| and Accessing Information | • Explain basic concepts, requirements, and terminologies of the Internet<br>• Apply the rules and regulations in the use of the internet.<br>• Using email<br>• Use the internet social network to communicate.<br>• Use Uniform Resource Locators (URLs) to access Information. Use search engines to access information<br>• Upload files to virtual drives and work on it. | OSI Reference Model, TCP/IP protocol stack, IPv4, IPv6, subnet Masking, MAC Address, Internet Services, Network Configuration and Troubleshooting, Wi-Fi, Broadband, Email Usage.<br>Rules and Regulations in the Use of the Internet: Spam-Unsolicited Emails, People's Privacy, Intellectual Property Rights, etc.<br><br>E-mail: Creating an Email Account, Sending, Accessing Email Messages, Attaching Documents to Email Messages,<br>Using the Internet: Social Networks to Communicate, Uniform Resource Locators (URLs) to Access Information, Using Search Engines, Downloading Information from the Internet. Transferring Information from the Internet to a Different Application<br><br>Intellectual Property Rights, ICT Policy, Software Piracy etc.<br><br>Using Cloud Space: Google Drive, Google Workspace, OneDrive, Dropbox, etc. | Practice |
| **Unit 8:** Emerging Technologies: Data analytics, Artificial intelligence, Machine learning | • Define Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML).<br>• Understand how data is collected, processed, and used for decision-making.<br>• Recognize the role of AI and ML in everyday life.<br>• Explain basic differences between AI | **Introduction to Data Analytics:** Define Data, Types of Data, Define Data Analytics, Basic Steps in Data Analytics, Applications of Data Analytics, Common Tools of Data Analytics ( Excel, Tableau, Power BI, Python, SQL, etc.).<br><br>**Artificial Intelligence (AI):** Define AI, Objectives of AI, AI capabilities, AI | Classroom Lectures, Lab and Hands-on Practice |

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| | and ML.<br>• Discuss benefits and challenges of these technologies.<br>• Recognize current trends and career opportunities in these fields. | applications (NLP, Image Recognition, Predictive Analytics, Recommender Systems, Robotics, etc.); **AI Tools**: OpenAI GPT, TensorFlow, PyTorch, Keras, Scikit-learn,etc.), AI in daily life (Voice assistants, facial recognition, GPS route suggestions, chatbots, etc.) .<br><br>**Machine Learning (ML):** Define Machine Learning (ML), How It Works, Classification of ML, applications of ML, and common ML platforms (Teachable Machine, TensorFlow, PyTorch, Microsoft Azure Machine Learning, etc.).<br><br>Relationship Between Data Analytics, AI, and ML<br><br>Benefits & Challenges of Data Analytics, AI, and ML. Future Trends & Career Paths. | |
| **Unit 9:** Introduction to Freelancing | The student will be able to:<br>• Define and explain the fundamental concept and significance of freelancing in the global context.<br><br>• Identify and categorize various online marketplaces and professional fields.<br><br>• Develop professional profiles, portfolios, and proposals suitable for | Introduction and Basic Concepts, Fields of Freelancing, Essential Skills and Preparation, Introduction to Freelancing Marketplaces, Creating Profile and Portfolio, Earning and Transaction Process, Building a Freelancing Career, Challenges and Solutions. | Classroom Lectures, Practical Demos, Assignments, Discussions |

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| | freelancing platforms.<br><br>• Demonstrate the use of online payment systems and financial transaction methods.<br><br>• Analyze and evaluate strategies for career development and effectively apply techniques to overcome freelancing challenges. | | |

**List of Books:**


Norton, P. (2008). *Introduction to computers* (9th ed.). The McGraw-Hill.

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft Office 2016 step by step*. Microsoft Press.

Hassan, N. A., & Hijazi, R. (2017). *Digital privacy and security using Windows: A practical guide*. Apress.

Maheshwari, A. (2024). *Data analytics made accessible*. Jay Cobb.

Russell, S. J., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Alpaydin, E. (2020). *Introduction to machine learning* (4th ed.). MIT Press.


| **Course Code** | **219904** | **Marks: 25** | **Credit: 1** |
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| **Course Title:** | **Lab: Information and Communication Technology** | | |

**Course Objectives:**

The main objective of the course is to teach the students' understanding and skills in using, and applying technology to solve problems and enhance various aspects of life and work. It includes assembling hardware, installing software, preparing PPT slides, and producing Word and Excel documents.

**Course Learning Outcomes:**

The student will be able to:

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| CLO 1 | Learn assembling hardware |
| CLO 2 | Prepare, edit and print Word documents and excel |
| CLO 3 | Prepare Power Point presentation. |
| CLO 4 | Access information from Internet, e-mail |
| CLO 5 | Installation of anti-virus software |
| CLO 6 | Data collection, Prediction, Pattern recognition, classification |
| CLO 7 | Exploring markets, Proposal writing, Simulation strategies |

**List of Experiments**

| CLO Addressed | Unit | Experiments | Teaching Learning Approach |
|---|---|---|---|
| CLO1 | 1-4 | • Assemble different hardware<br>• Install different software<br>• Operate the computer - Drive, folder and file management<br>• Maintenance | Lab and Hands-on Practice |
| CLO 2 | 3 | **Word**<br>• Prepare a Word document on a specific topic (e.g, routine, question paper, CV, reports, applications)<br>• Formatting the document (Alignment, table, border, watermark, etc.), e.g., newspaper article, academic report, or documentation used in daily life, book, poster<br>• Print documents with different paper and printers | Lab and Hands-on Practice |
| CLO 2 | 4 | • **Excel**<br>• Prepare a grade sheet<br>• Prepare a family expenditure<br>• Prepare a business expenditure report<br>• Prepare payroll management, with a report<br>• Create graphs on the given data<br>• Print Excel files | Lab and Hands-on Practice |

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| CLO 3 | 5 | **Power point**<br>• Prepare an academic presentation on a specific topic.<br>• Formatting the slides & using different tools.<br>• Apply animation and transition<br>• Print PPT files in different modes: Hand note, Slides shorter, outline | Lab and Hands-on Practice |
| CLO4 | 6 | • Install antivirus software, e.g., Norton Antivirus, McAfee, Kaspersky, Avast. | Lab and Hands-on Practice |
| CLO5 | 7 | 1. Use of email<br>2. Access information from the internet, use a search engine.<br>3. Use of virtual drive for collaboration<br>4. Google Meet | Lab and Hands-on Practice |
| CLO6 | 8 | • Data Collection and Visualization (Excel)<br>• Simple Prediction (Excel)<br>• Pattern recognition (Teachable Machine)<br>• Classification (Teachable Machine) | Lab and Hands-on Practice |
| CLO7 | 9 | **Free lancing**<br>• Marketplace exploration and portfolio creation<br>• Proposal writing<br>• Payment simulation | Lab, Hands-on, Group Work, Discussion |

**Instructions for the Lab:**

1. Lab sessions should be hands-on, with students engaging in tasks like assembling hardware, installing software, and using office tools.
2. Software like Microsoft Word, Excel, PowerPoint, and Teachable Machine will be used for document creation, data analysis, and basic machine learning.
3. Labs also include antivirus installation, email use, and online collaboration via Google Meet and Drive, emphasizing teamwork and problem-solving.

**Instructions for the Lab exam:**

1. The lab exam is individual. Students must answer two out of seven questions.
2. Marks distribution: Experiment (20 marks, 10 for each) for practical tasks, Viva-Voce (5 marks) for oral assessment. Total: 25 marks.

**List of Books:**

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft office 2016 step by step*. Microsoft Press